technologies such as AI-driven flight scheduling, satellite navigation, and digital airspace mapping, IAM systems aim to maintain safe separation between drones and manned aircraft while allowing simultaneous operations in dense airspaces.

Collaboration between civil aviation regulators and defense agencies is also essential to define segregated air corridors, priority zones, and dynamic access protocols. Some nations have adopted the concept of flexible use of airspace (FUA), wherein airspace is dynamically allocated to either civilian or military users based on real-time demand and operational necessity. This model enables more efficient use of limited airspace resources without compromising national security or public safety.

However, successful airspace integration faces challenges related to data sharing, interoperability, and jurisdictional boundaries. Defense agencies often restrict information about military operations, while civilian regulators prioritize transparency and safety. Bridging this gap requires secure data exchange protocols, standardized communication systems, and encrypted identification methods that allow mutual awareness without exposing sensitive mission data.

In the future, the convergence of AI-assisted UTM systems, digital twin simulations, and real-time airspace analytics will play a pivotal role in enabling smooth civilian-military airspace coexistence. A unified, adaptive, and technology-driven airspace management ecosystem will not only enhance operational safety and efficiency but also establish a foundation for next-generation aerial mobility, where drones of diverse functions, commercial, defense, and humanitarian can operate collaboratively in a shared sky.

7.4.3 Managing air traffic and ensuring safety in densely